p-values — Explained Simply

4 min read · Apr 18, 2026

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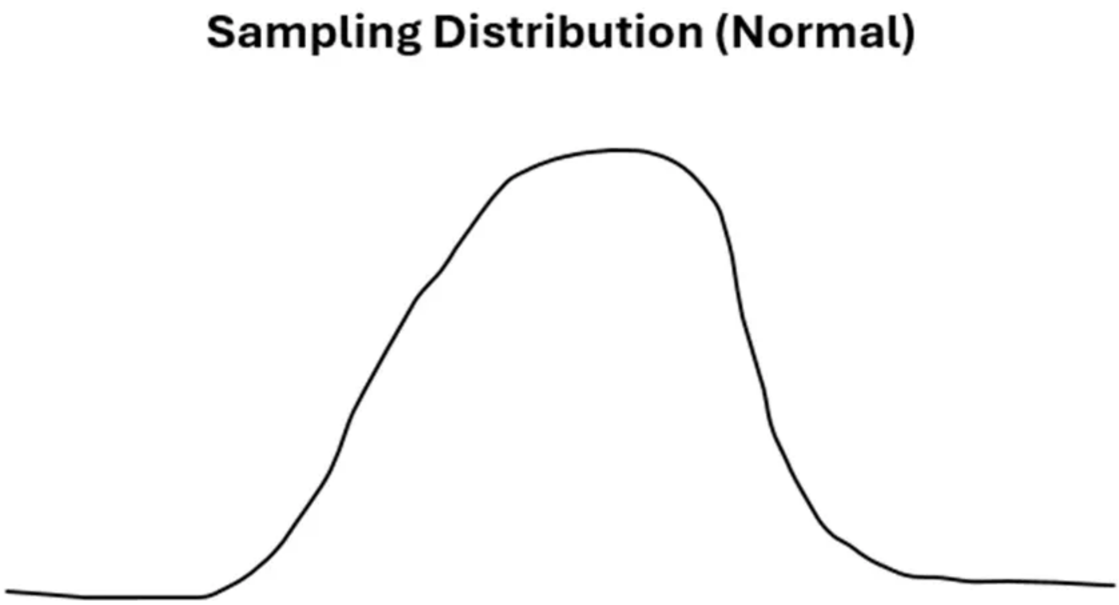
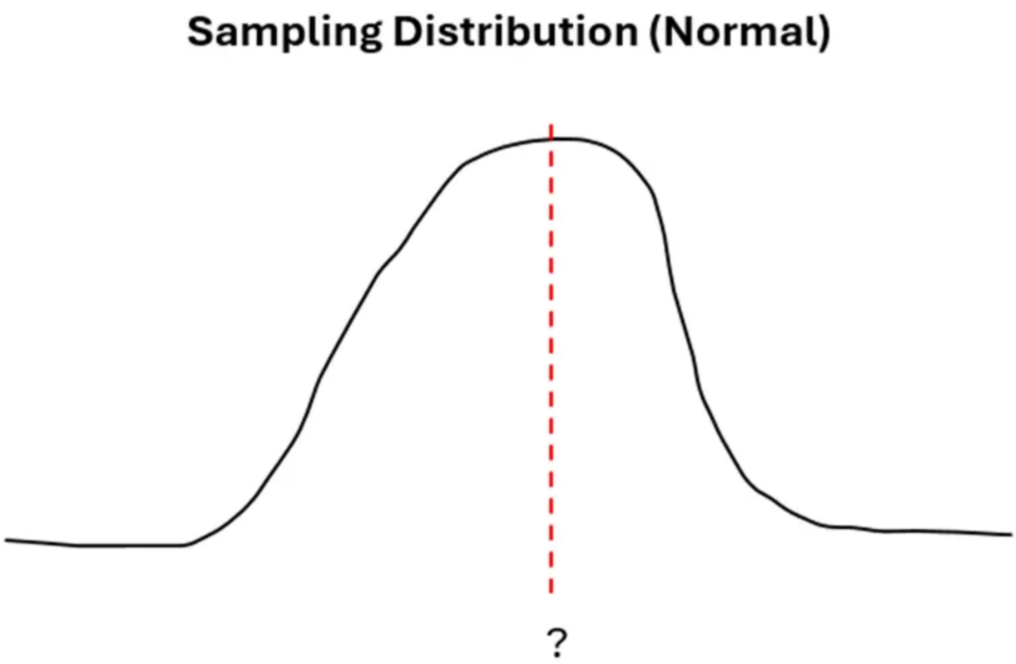
What exactly are p-values, and why do we want them to be lower than 0.05? To understand p-values — you NEED to understand the sampling distribution. There's no other way around it. So if you need a reminder on what a sampling distribution is, refer to the link above!

So remember — a sampling distribution is a HYPOTHETICAL construct describing how the SAMPLE MEANS would vary if you collected many many samples from the population. Because of the central limit theorem, we can confidently say that the shape of the sampling distribution WILL BE NORMAL — even if we don't make any assumptions about the population distribution.

A sampling distribution will look like that:



As of now, we can only confidently state the SHAPE of the sampling distribution. The question is — what is the MEAN of this sampling distribution?



It actually is the mean of the POPULATION DISTRIBUTION — because the mean of EVERY sample is most likely to congregate around the this population mean *(the precise terminology is that the sample mean is an unbiased estimator of the population mean, but let's not use fancy jargon over here heh).*

Side note: If you're wondering why this is true, it's just fundamental logic already — randomly sampling from a population distribution, the mean of the sample WILL be most likely to be the population mean as well. You can just do a simple thought experiment to verify.

And now we face a problem — because if we DON'T want to make any assumptions about the population distribution — how can we construct this sampling distribution?

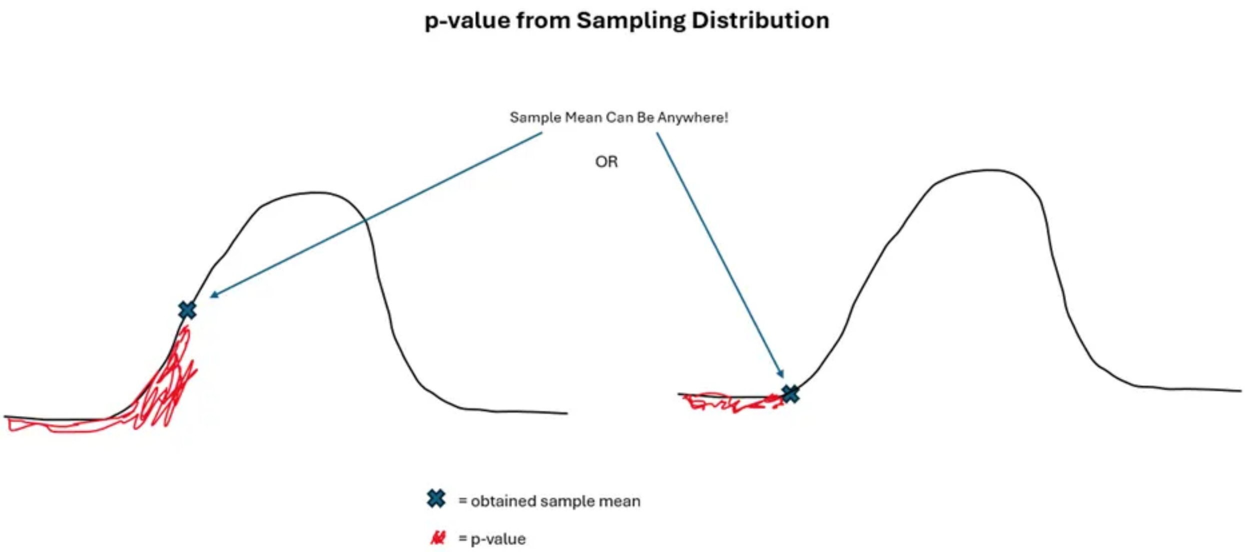
p-value starts with ASSUMPTION that H0 is True

Which leads us to the first golden rule of the p-value — it rests on the ASSUMPTION that the null hypothesis is true. What is the null hypothesis? It is a PRESUMED VALUE for the population mean — often written like this:

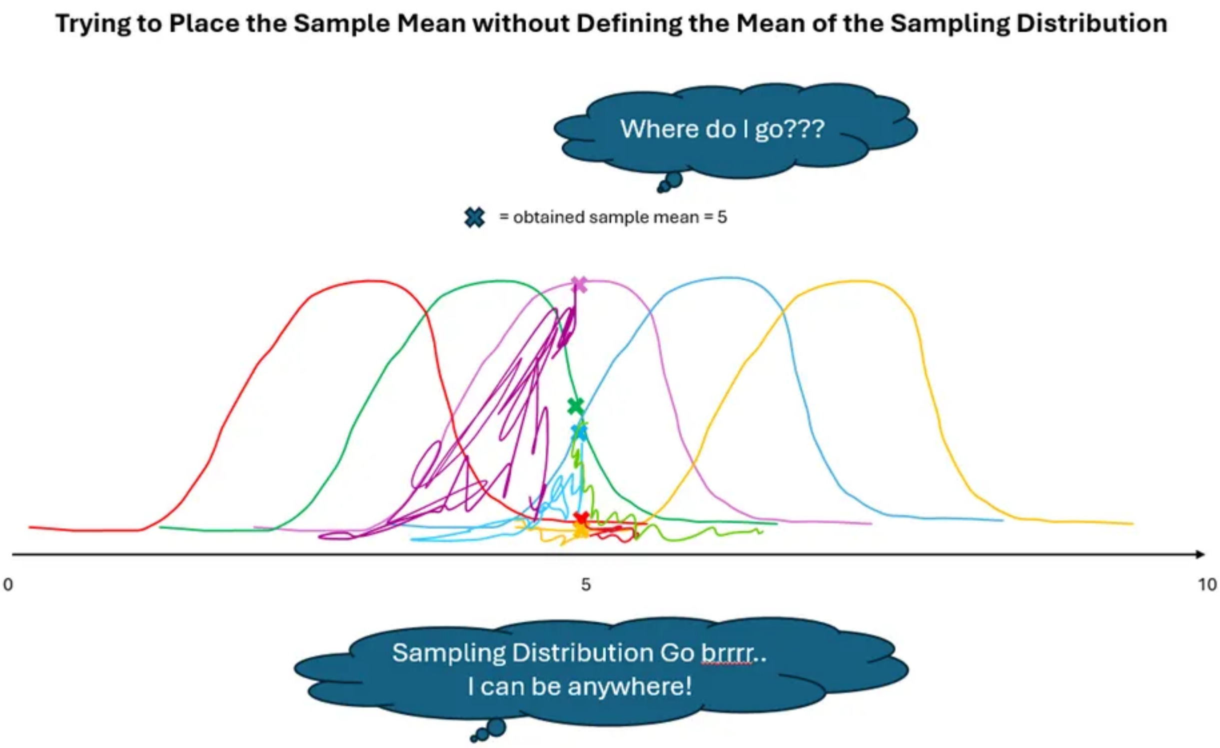
$$H_0 : \mu = 5$$

Why is this important? Because you CAN'T construct the sampling distribution without knowing the value of the population mean — and since by definition you DON'T KNOW the population mean, you have to PRESUME a value for it first!

And only with the sampling distribution defined — can you compute a p-value. Because a p-value represents the *probability of obtaining a sample mean as least as extreme as your obtained sample* ASSUMING THE NULL HYPOTHESIS IS TRUE- which is represented by this red area.



In order to compute a p-value — you need to PLACE your sample mean within the sampling distribution. And if you don't define the population mean of the sampling distribution — you CAN'T place your sample on this sampling distribution — and you can't compute a p-value!



The p-value — without properly defining the mean of the sampling distribution — can be ANY ONE of the shaded areas above. So you NEED to presume a value for the population mean FIRST (null hypothesis) — BEFORE you can compute a p-value!

Why do we want p < 0.05?

So now that you know what a p-value is — what's the deal with 0.05? Oftentimes — we're looking for evidence AGAINST the null hypothesis. Remember that we always want to DISPROVE the null hypothesis — in order to support that fact the alternative hypotheses are plausible! (NOT CORRECT — be precise here!).

Oftentimes — this comes in the form of research questions like: I want to prove that "my intervention has improved crop yield", "the average height has increased compared to 5 years ago", or "diabetes is more prevalent compared to the 1990s". In all of these instance — you are trying to collect evidence that IT IS UNLIKELY that your sample would have been obtained IF the null hypothesis were true (presumed population mean). As such, more likely than not, the population HAS CHANGED.

This is why the p-value ITSELF is not yet a decision tool — it merely tells you the probability of obtaining a sample mean at least as extreme as your sample mean assuming the null hypothesis is true (so clunky!). This is just a probability — it needs to be compared to a THREHOLD VALUE (alpha, often 0.05) before a decision can be made.

What is this decision? It is the decision that "okay — this is unlikely *enough*. If the null hypothesis WERE true, we probably would not have obtained this sample mean. This sample mean is extreme enough to constitute EVIDENCE AGAINST THE NULL HYPOTHESIS. Let's take it that the null hypothesis is thus NOT true, BECAUSE of my collected sample.

Understand now? 😉

Side note: This is part of a broader series on statistics fundamentals, which goes into more detail on related topics.

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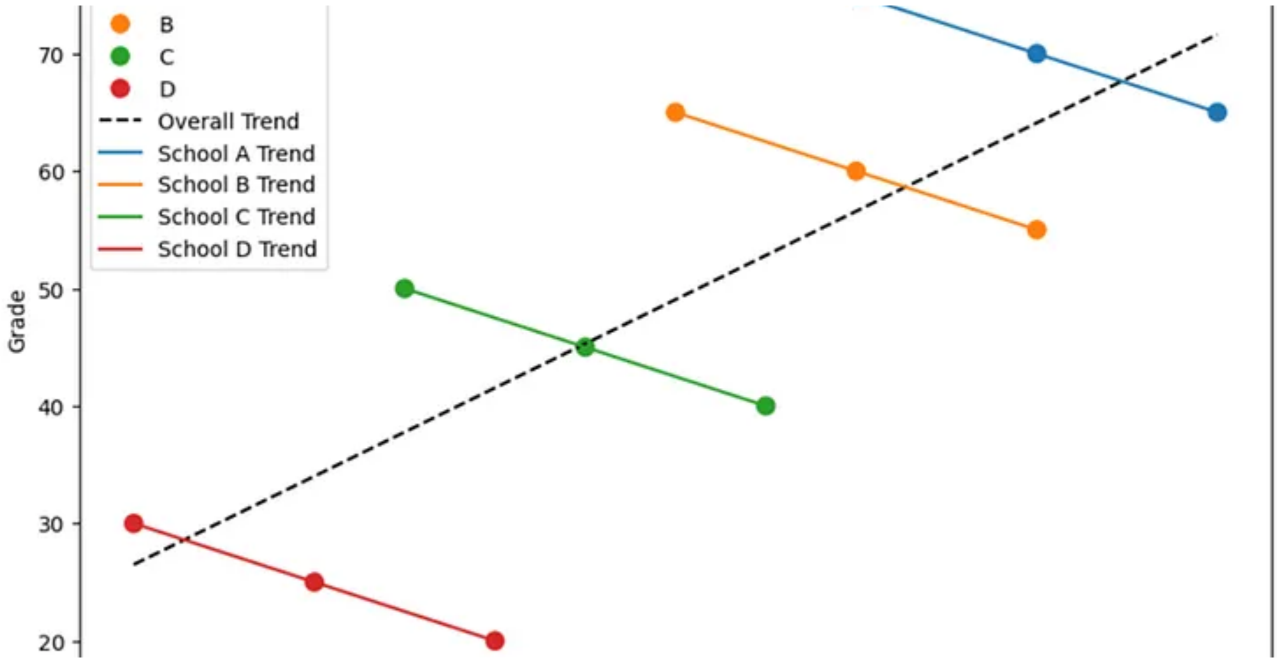

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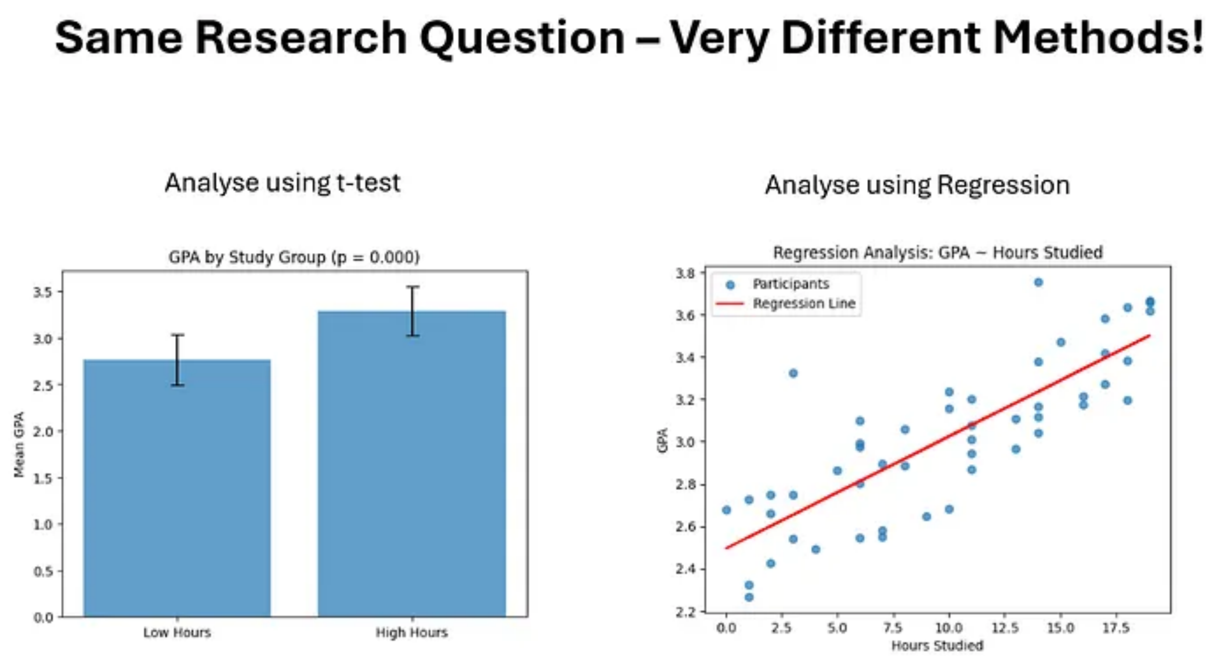

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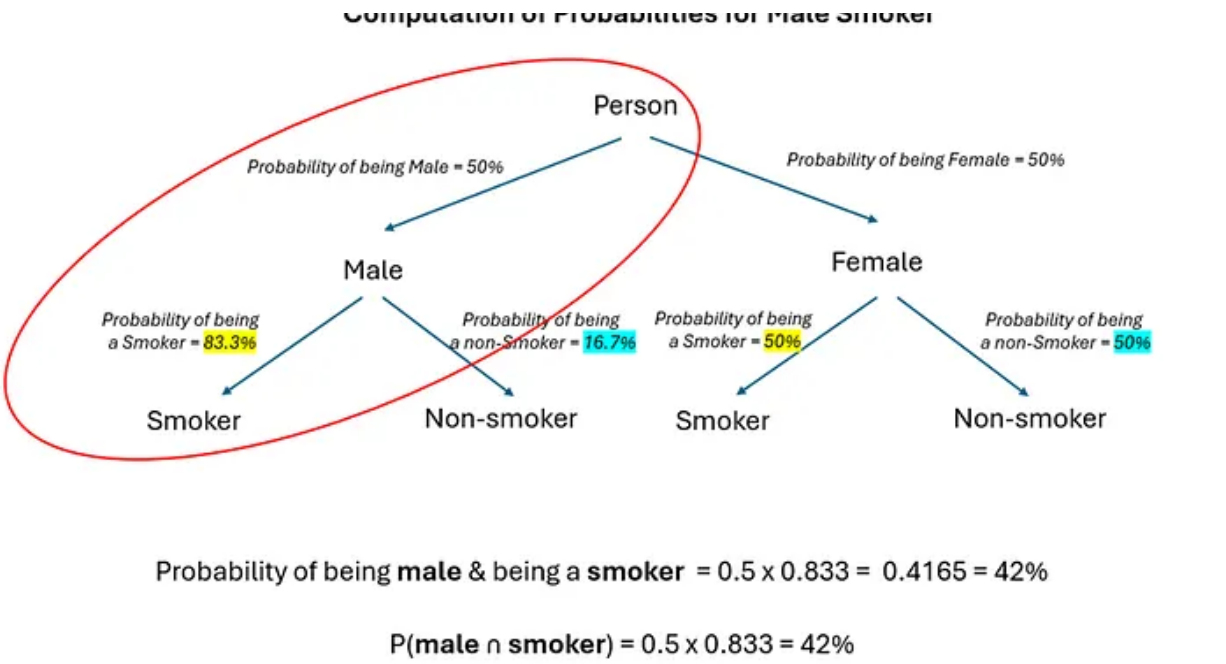

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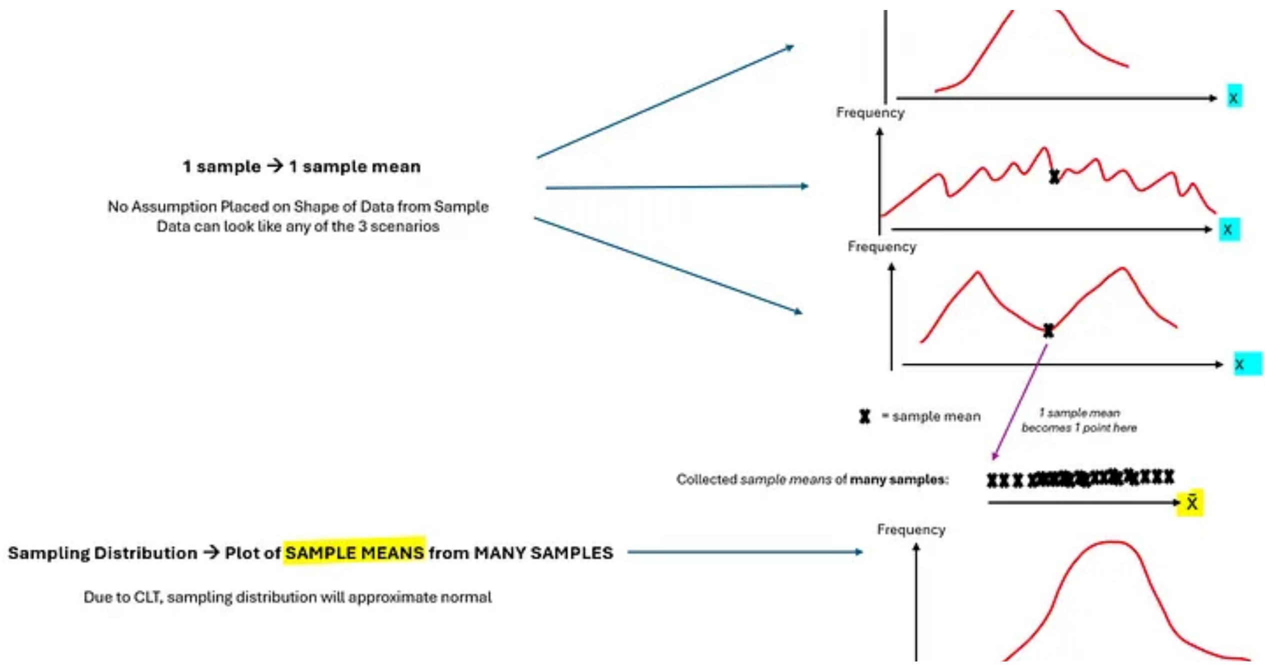

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